

◆ **Step 1 — Pivot in column 1**

Swap $R_1 \leftrightarrow R_2$:

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 2 \\ -1 & 0 & 0 & 1 & -3 \\ 0 & -1 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 & 6 \end{array} \right].$$

Eliminate below:

- $R_2 \leftarrow R_2 + R_1 = [0, -1, 0, 1 | -1]$.

Matrix becomes:

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 2 \\ 0 & -1 & 0 & 1 & -1 \\ 0 & -1 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 & 6 \end{array} \right].$$

Eliminate col 2:

- $R_1 \leftarrow R_1 + R_2 = [1, 0, 0, -1 | 3]$.
- $R_3 \leftarrow R_3 + R_2 = [0, 0, 1, -1 | 6]$.

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 3 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 6 \\ 0 & 0 & 1 & -1 & 6 \end{array} \right].$$

◆ **Step 3 — Pivot in column 3**

Row 3 pivot is already 1.

Eliminate duplicate: $R_4 \leftarrow R_4 - R_3 = [0, 0, 0, 0 | 0]$.

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 3 \\ 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

◆ Step 4 — Put into RREF

Right now each pivot row has a -1 in col 4.

We treat x_4 as **free variable** = t .

To write in RREF, isolate pivot variables:

- Row 1: $x_1 - x_4 = 3 \implies x_1 = 3 + x_4$.
- Row 2: $x_2 - x_4 = 1 \implies x_2 = 1 + x_4$.
- Row 3: $x_3 - x_4 = 6 \implies x_3 = 6 + x_4$.

So the RREF augmented matrix looks like:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & t \end{array} \right],$$

where t is a free parameter.

👉 The system has infinitely many solutions because the coefficient matrix is **singular**. The free variable t corresponds to an arbitrary circulation around the loop.

Would you like me to also **illustrate this with a step-by-step substitution tree** (showing how each variable depends on t)?